

PROJECT ROLES & RESOURCE ALLOCATION

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Project Developer Profile

The OptiCrop project was developed entirely as an individual project. The developer executed all roles across the software development lifecycle:

Development Role	Responsibilities Executed	Key Deliverables Completed
Product Lead	Requirements elicitation, planning, and documentation.	SRs, empathy mapping, and user journey maps.
Member A - Lead Presenter	Delivers introduction, problem framing, closing roadmap & Q&A.	
ML Engineer	Data preprocessing, algorithm selection, and training.	Trained model pipelines and evaluation plots.
Member B - ML Presenter	Explains dataset, model selection, accuracy metrics, confusion matrix.	
Backend Developer	Flask server controller logic, route validation, database logic.	api.py, logger, predictions.csv, health check routes.
Member D - QA Presenter	Explains dataset, model selection, accuracy metrics, confusion matrix.	
UI/UX Designer	Glassmorphic dashboard style guidelines, responsive layout.	index.html, routes.html, dashboard.html templates.

Performs live web-app demo: form input -> crop result -> dashboard.

Member C - App Presenter

Member D - QA Presenter

Performs live web-app demo: form input -> crop result -> dashboard.

Runs pytest live, shows CSV log, explains test coverage.

Member D - QA Presenter

Runs pytest live, shows CSV log, explains test coverage.